ENUNCIADO — PROBLEMA 1

Numerical Analysis Preliminary Examination questions
January 2018

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1. Let $f \in C^{1,\alpha}[x_0, x_1]$ be a C^1 function whose derivative f' is Hölder α with $0 < \alpha \leq 1$. Let p be the linear interpolant of f at x_0 and x_1. If $w(t)$ denotes the modulus of continuity of f', show the error estimate

$$|f(x) - p(x)| \leq hw(h) \quad \forall x_0 < x < x_1,$$

where $h = x_1 - x_0$.
Hint: Express the differences $f(x) - f(x_0)$ and $f(x_1) - f(x_0)$, which appear when writing $f(x) - p(x)$, in integral form and then combine them. Also recall from the definition of modulus of continuity which implies $|f(x) - f(y)| \leq w(|x - y|)$.

2. Gauss Quadrature

   (a) Derive the two point Gaussian integration formula for

$$I(f) = \int_{-1}^{1} f(x)\, dx.$$

   (b) Use this formula to approximate $I(x^2)$.

3. Let A be diagonalizable, and consider the power iteration sequence $w_{k+1} = Aw_k$ for an initial vector w_0. Find the limit $\lim_{k\to\infty} \frac{w_k}{||w_k||}$ and show how this limit depends on the spectrum of A and the initial vector w_0.

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5. Consider a minimization problem $f(x) = \frac{1}{2}x^T Ax - b^T x$, $x \in \mathbb{R}^n$ and A is symmetric positive definite.

   (a) Prove that $f(x)$ is minimized when $Ax = b$.

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   (c) Perform one step of both Jacobi and Gauss-Seidel iterative methods to solve $Ax = b$ and compare the two methods.

1

SOLUCIÓN — PROBLEMA 1

Problema 1 — Error de interpolación lineal vía módulo de continuidad

Sea $h = x_1 - x_0$, $\theta = \dfrac{x - x_0}{h} \in [0,1]$ para $x \in [x_0, x_1]$, de modo que $x - x_0 = \theta h$ y $x_1 - x = (1-\theta)h$. El interpolante lineal es $p(x) = f(x_0) + \theta[f(x_1) - f(x_0)]$.

Paso 1 (reescritura algebraica).

$$f(x) - p(x) = [f(x) - f(x_0)] - \theta[f(x_1) - f(x_0)] = (1-\theta)[f(x) - f(x_0)] - \theta[f(x_1) - f(x)],$$

(sumando y restando $\theta[f(x) - f(x_0)]$, o directamente verificando la identidad algebraica).

Paso 2 (forma integral, centrada en x). Por el teorema fundamental del cálculo,

$$f(x) - f(x_0) = \int_{x_0}^{x} f'(t)\,dt = \int_{x_0}^{x} \big[f'(t) - f'(x)\big]\,dt + f'(x)(x - x_0),$$
$$f(x_1) - f(x) = \int_{x}^{x_1} f'(t)\,dt = \int_{x}^{x_1} \big[f'(t) - f'(x)\big]\,dt + f'(x)(x_1 - x).$$

Sustituyendo en el Paso 1, los términos con $f'(x)$ son $(1-\theta)f'(x)(x - x_0) - \theta f'(x)(x_1 - x) = (1-\theta)\theta h \;\, f'(x) - \theta(1-\theta)h \;\, f'(x) = 0$ (porque $x - x_0 = \theta h$ y $x_1 - x = (1-\theta)h$). Por lo tanto

$$f(x) - p(x) = (1-\theta)\int_{x_0}^{x} \big[f'(t) - f'(x)\big]\,dt - \theta\int_{x}^{x_1} \big[f'(t) - f'(x)\big]\,dt.$$

Paso 3 (acotación con el módulo de continuidad). Por definición de w, $|f'(t) - f'(x)| \le w(|t - x|)$ para todo $t, x \in [x_0, x_1]$. Como $t, x \in [x_0, x_1]$ implica $|t - x| \le h$, y w es no decreciente, $w(|t - x|) \le w(h)$. Entonces, por la desigualdad triangular,

$$|f(x) - p(x)| \le (1-\theta)\int_{x_0}^{x} w(h)\,dt + \theta\int_{x}^{x_1} w(h)\,dt = (1-\theta)(x - x_0)w(h) + \theta(x_1 - x)w(h).$$

Sustituyendo $x - x_0 = \theta h$, $x_1 - x = (1-\theta)h$:

$$|f(x) - p(x)| \le \big[(1-\theta)\theta h + \theta(1-\theta)h\big]w(h) = 2\theta(1-\theta)\,h\,w(h).$$

Como $\theta(1-\theta) \le \frac{1}{4}$ para $\theta \in [0,1]$ (máximo en $\theta = 1/2$),

$$|f(x) - p(x)| \le 2 \cdot \frac{1}{4} \cdot h\,w(h) = \frac{h}{2}w(h) \le h\,w(h) \qquad \forall x_0 < x < x_1. \;\blacksquare$$

(Obtenemos de hecho la cota más fuerte $\frac{1}{2}hw(h)$, que implica la solicitada $hw(h)$.)

ENUNCIADO — PROBLEMA 2

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SOLUCIÓN — PROBLEMA 2

Problema 2 — Cuadratura gaussiana de 2 puntos en $[-1, 1]$

(a) Por simetría del intervalo y del peso ($w \equiv 1$, par), buscamos nodos $\pm\xi$ y pesos iguales $w_1 = w_2 = w$ (la simetría hace automática la exactitud para f impares). Exactitud para $f = 1$: $2w=\int_{-1}^1dx=2 \Rightarrow w=1$. Exactitud para $f=x^2$:

$$2w\xi^2 = \int_{-1}^{1} x^2 dx = \frac{2}{3} \implies \xi^2 = \frac{1}{3} \implies \xi = \frac{1}{\sqrt{3}}.$$

$$\boxed{I(f) \approx f\left(-\tfrac{1}{\sqrt{3}}\right) + f\left(\tfrac{1}{\sqrt{3}}\right).}$$

Esta regla es exacta para todo polinomio de grado ≤ 3 (2 nodos $\Rightarrow$ grado $2 \cdot 2 - 1 = 3$), en particular para $1, x, x^2, x^3$.

(b) $I(x^2)=\left(-\tfrac1{\sqrt3}\right)^2+\left(\tfrac1{\sqrt3}\right)^2 =\tfrac13+\tfrac13=\tfrac23$, que coincide exactamente con el valor exacto $\int_{-1}^{1} x^2 dx = \frac{2}{3}$ (como debía ser, por la exactitud hasta grado 3).

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SOLUCIÓN — PROBLEMA 3

Problema 3 — Límite de la iteración de potencias

Sea A diagonalizable con eigenpares (λ_i, v_i), $i = 1, \dots, n$, ordenados de modo que existe un **eigenvalor dominante** simple, es decir $|\lambda_1| > |\lambda_2| \geq \cdots \geq |\lambda_n|$. Escribimos el vector inicial en la base de eigenvectores: $w_0 = \sum_{i=1}^{n} c_i v_i$.

Cálculo. Como $w_k = A^k w_0$,

$$w_k = A^k w_0 = \sum_{i=1}^{n} c_i \lambda_i^k v_i = \lambda_1^k \left[c_1 v_1 + \sum_{i=2}^{n} c_i \left(\frac{\lambda_i}{\lambda_1} \right)^k v_i \right].$$

Como $|\lambda_i / \lambda_1| < 1$ para $i \geq 2$, el término entre corchetes tiende, cuando $k \to \infty$, a $c_1 v_1$ **siempre que** $c_1 \neq 0$ (es decir, que w_0 tenga componente no nula en la dirección del eigenvector dominante v_1 — esto ocurre para "casi todo" w_0, salvo los que caen exactamente en el subespacio invariante generado por $v_2, \dots, v_n$).

Normalizando:

$$\frac{w_k}{\|w_k\|} = \frac{\lambda_1^k [c_1 v_1 + \varepsilon_k]}{|\lambda_1|^k \|c_1 v_1 + \varepsilon_k\|} = \operatorname{sign}(\lambda_1)^k \cdot \frac{c_1 v_1 + \varepsilon_k}{\|c_1 v_1 + \varepsilon_k\|} \xrightarrow[k \to \infty]{} \operatorname{sign}(\lambda_1)^k \cdot \operatorname{sign}(c_1) \frac{v_1}{\|v_1\|},$$

donde $\varepsilon_k = \sum_{i>2} c_i (\lambda_i / \lambda_1)^k v_i \to 0$.

Conclusión.

- Si $\lambda_1 > 0$: $\displaystyle\lim_{k\to\infty}\frac{w_k}{\|w_k\|}

=\operatorname{sign}(c_1)\,\frac{v_1}{\|v_1\|}$, un vector fijo (el eigenvector dominante normalizado, con signo determinado por la proyección inicial c_1).

- Si $\lambda_1 < 0$: el factor $\operatorname{sign}(\lambda_1)^k = (-1)^k$

alterna en cada paso, así que $w_k / \|w_k\|$ **no converge** como sucesión, sino que oscila entre los dos límites antipodales $\pm v_1 / \|v_1\|$ a lo largo de las subsucesiones de índice par/impar (equivalentemente, converge en el sentido "proyectivo": la *recta* generada por w_k converge a la recta generada por v_1).

Dependencia del espectro y de w_0: el límite existe (en el sentido de dirección) si y solo si (i) hay un único eigenvalor de módulo máximo (dominancia estricta) y (ii) w_0 tiene componente no nula en su eigenespacio ($c_1 \neq 0$); la velocidad de convergencia es geométrica, con razón asintótica $|\lambda_2 / \lambda_1| < 1$. ■

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SOLUCIÓN — PROBLEMA 4

Problema 4 — Inversión de una matriz $N \times N$ en $O(N^3)$

Algoritmo.

1. Calcular la factorización LU (con pivoteo parcial) de A: $PA = LU$.
2. Para cada $i = 1, \dots, N$, resolver el sistema $Ax_i = e_i$ (donde e_i es la

i-ésima columna de la identidad) usando la factorización ya calculada: resolver $Ly_i = Pe_i$ por sustitución hacia adelante, luego $Ux_i = y_i$ por sustitución hacia atrás.

1. Ensamblar $A^{-1} = [x_1 \ x_2 \ \cdots \ x_N]$.

Justificación de la complejidad.

- *Factorización LU:* la eliminación gaussiana triangulariza A en

$\Theta(N^3)$ operaciones. Concretamente, en el paso k ($k = 1, \dots, N-1$) se actualiza una submatriz de tamaño $(N-k) \times (N-k)$, con $\sim (N-k)^2$ multiplicaciones; sumando,

$$\sum_{k=1}^{N-1} (N-k)^2 = \sum_{j=1}^{N-1} j^2 = \frac{(N-1)N(2N-1)}{6} = \Theta(N^3).$$

- *Sustituciones triangulares:* por el conteo del Problema 6 del examen de

enero de 2018 (idéntico razonamiento), resolver un sistema triangular $N \times N$ cuesta $\Theta(N^2)$ operaciones (del orden $N^2/2$ multiplicaciones más N divisiones). Como hay que resolver **dos** sistemas triangulares ($Ly_i = \cdot$ y $Ux_i = \cdot$) para cada una de las N columnas e_i, el costo total de esta etapa es

$$N \cdot \Theta(N^2) = \Theta(N^3).$$

Total: $\Theta(N^3) + \Theta(N^3) = \Theta(N^3)$, es decir $O(N^3)$. (La factorización LU se calcula **una sola vez**, y se reutiliza para las N resoluciones triangulares — de ahí que el costo total siga siendo cúbico y no $O(N^4)$, que sería el costo ingenuo de resolver N sistemas por eliminación gaussiana completa desde cero cada vez.) ■

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6. Consider the Newton-Raphson method: $x_{n+1} = x_n - \dfrac{f(x_n)}{f'(x_n)}$

   (a) Use the Newton-Raphson method to establish the following iterative formula $x_{n+1} = \dfrac{x_n^2 + 16}{2x_n}$ for $n > 0$ to find the square-root of 16.

   (b) Show that if the sequence $\{x_n\}$ converges to the limit $L = 4$, then the convergence rate is quadratic. (Hint: Consider the error $e_n = x_n - 4$.)

7. Consider the two-point boundary value problem:

$$-u'' + u = \sin(x) \quad \text{in } (0,1), \quad \text{with } u(0) = u(1) = 0. \tag{1}$$

   (a) Write the finite difference approximation of (1) using *centered* differences on a uniform mesh with meshsize h. Write the matrix of the system and examine how the equations would change if $u(0) = \alpha \neq 0$.

   (b) Set up the finite element approximation for this problem, based on piecewise linear elements in equidistant points. State the convergence rate in an appropriate norm.

SOLUCIÓN — PROBLEMA 5

Problema 5 — Minimización cuadrática y métodos iterativos

(a) $f(x) = \frac{1}{2}x^TAx - b^Tx,\ A = A^T \succ 0$. El gradiente es $\nabla f(x) = Ax - b$ y el Hessiano es $\nabla^2 f(x) = A \succ 0$ (constante). Sea $x^* = A^{-1}b$ (el único punto donde $\nabla f = 0$, ya que A es invertible). Como f es cuadrática, para cualquier $y \in \mathbb{R}^n$ vale la expansión **exacta** de Taylor:

$$f(y) = f(x^*) + \nabla f(x^*)^T(y - x^*) + \frac{1}{2}(y - x^*)^TA(y - x^*) = f(x^*) + \frac{1}{2}(y - x^*)^TA(y - x^*),$$

(el término lineal se anula porque $\nabla f(x^*) = Ax^* - b = 0$). Como $A \succ 0$, $(y - x^*)^TA(y - x^*)$ > 0 para todo $y \neq x^*$, así que

$$f(y) > f(x^*) \quad \text{para todo } y \neq x^*.$$

Luego x^* es el **único** minimizador global de f, y satisface $Ax^* = b$. ■

(b) Un paso de descenso más pronunciado desde x_0: la dirección de descenso es $d_0 = -\nabla f(x_0) = b - Ax_0 =: r_0$ (el residuo). Búsqueda de línea exacta: minimizar $\varphi(\alpha) = f(x_0 + \alpha r_0)$ en α.

$$\varphi(\alpha) = f(x_0) + \alpha\, \nabla f(x_0)^T r_0 + \frac{1}{2}\alpha^2 r_0^T A r_0 = f(x_0) - \alpha\|r_0\|^2 + \frac{1}{2}\alpha^2\, r_0^T A r_0,$$

usando $\nabla f(x_0)^T r_0 = -r_0^T r_0 = -\|r_0\|^2$. Como $r_0^T A r_0 > 0$ (a menos que $r_0 = 0$, en cuyo caso $x_0 = x^*$ ya), φ es una parábola convexa en α; su mínimo se alcanza donde $\varphi'(\alpha) = 0$:

$$\varphi'(\alpha) = -\|r_0\|^2 + \alpha\, r_0^T A r_0 = 0 \implies \alpha_0 = \frac{\|r_0\|^2}{r_0^T A r_0}.$$

$$\boxed{x_1 = x_0 + \alpha_0 r_0 = x_0 + \frac{r_0^T r_0}{r_0^T A r_0}\, r_0, \qquad r_0 = b - Ax_0.}$$

(c) Escribimos $A = D - L - U$ con D diagonal, $-L$ la parte estrictamente triangular inferior y $-U$ la parte estrictamente triangular superior de A.

Un paso de Jacobi (todas las componentes usan solo $x^{(0)}$):

$$x_i^{(1)} = \frac{1}{a_{ii}}\left(b_i - \sum_{j\neq i} a_{ij}x_j^{(0)}\right), \qquad i = 1, \dots, n, \quad \text{simultáneamente, es decir } x^{(1)} = D^{-1}(b + (L + U)x^{(0)}).$$

Un paso de Gauss-Seidel (usa inmediatamente las componentes ya actualizadas dentro del mismo barrido, en orden $i = 1, \dots, n$):

$$x_i^{(1)} = \frac{1}{a_{ii}}\left(b_i - \sum_{j<i} a_{ij}x_j^{(1)} - \sum_{j>i} a_{ij}x_j^{(0)}\right), \qquad x^{(1)} = (D - L)^{-1}(b + Ux^{(0)}).$$

Comparación. Jacobi actualiza todas las componentes "en paralelo" usando únicamente información de la iteración anterior — es fácilmente paralelizable pero típicamente converge más lentamente. Gauss-Seidel usa la información más reciente tan pronto está disponible (actualización secuencial/in-place) — no es tan fácil de paralelizar, pero converge en general más rápido (radio espectral menor). Para A simétrica definida positiva (como aquí), **Gauss-Seidel siempre converge** (teorema clásico:

$\rho\left((D - L)^{-1}U\right) < 1$ si $A \succ 0$), mientras que la convergencia de Jacobi para SPD **no está garantizada en general** (requiere, por ejemplo, dominancia diagonal adicional).

ENUNCIADO — PROBLEMA 6

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SOLUCIÓN — PROBLEMA 6

Problema 6 — Newton-Raphson para $\sqrt{16}$

(a) $f(x) = x^2 - 16$, $f'(x) = 2x$:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = x_n - \frac{x_n^2 - 16}{2x_n} = \frac{2x_n^2 - x_n^2 + 16}{2x_n} = \frac{x_n^2 + 16}{2x_n}.$$

(b) Sea $e_n = x_n - 4$. Entonces

$$x_{n+1} - 4 = \frac{x_n^2 + 16}{2x_n} - 4 = \frac{x_n^2 - 8x_n + 16}{2x_n} = \frac{(x_n - 4)^2}{2x_n} = \frac{e_n^2}{2x_n}.$$

Si $x_n \to 4$ (y $x_n \neq 0$ en el proceso), entonces

$$\frac{e_{n+1}}{e_n^2} = \frac{1}{2x_n} \xrightarrow[n\to\infty]{} \frac{1}{8},$$

un límite finito y no nulo: la convergencia es **cuadrática** (orden 2), con constante asintótica de error $C = 1/8$. ■

ENUNCIADO — PROBLEMA 7

6. Consider the Newton-Raphson method: $x_{n+1} = x_n - \dfrac{f(x_n)}{f'(x_n)}$

   (a) Use the Newton-Raphson method to establish the following iterative formula $x_{n+1} = \dfrac{x_n^2 + 16}{2x_n}$ for $n > 0$ to find the square-root of 16.

   (b) Show that if the sequence $\{x_n\}$ converges to the limit $L = 4$, then the convergence rate is quadratic. (Hint: Consider the error $e_n = x_n - 4$.)

7. Consider the two-point boundary value problem:

$$-u'' + u = \sin(x) \quad \text{in } (0,1), \quad \text{with } u(0) = u(1) = 0. \tag{1}$$

   (a) Write the finite difference approximation of (1) using *centered* differences on a uniform mesh with meshsize h. Write the matrix of the system and examine how the equations would change if $u(0) = \alpha \neq 0$.

   (b) Set up the finite element approximation for this problem, based on piecewise linear elements in equidistant points. State the convergence rate in an appropriate norm.

SOLUCIÓN — PROBLEMA 7

Problema 7 — BVP $-u'' + u = \sin(x)$, $u(0) = u(1) = 0$

(a) Diferencias finitas. Malla uniforme $x_i = ih$, $i = 0, \ldots, N$, $h = 1/N$, nodos interiores $i = 1, \ldots, N-1$, con $u_0 = u(0) = 0$, $u_N = u(1) = 0$ conocidos. Diferencia central para u'':

$$-\frac{u_{i-1} - 2u_i + u_{i+1}}{h^2} + u_i = \sin(x_i) \implies -u_{i-1} + (2+h^2)u_i - u_{i+1} = h^2 \sin(x_i), \quad i = 1, \ldots, N-1.$$

En forma matricial, $A_h\mathbf{u} = \mathbf{f}$ con $\mathbf{u} = (u_1, \ldots, u_{N-1})^T$,

$$A_h = \begin{pmatrix} 2+h^2 & -1 & & \\ -1 & 2+h^2 & -1 & \\ & \ddots & \ddots & \ddots \\ & & -1 & 2+h^2 \end{pmatrix}, \qquad f_i = h^2 \sin(x_i) \ (i = 2, \ldots, N-2 \text{ sin cambios}).$$

A_h es tridiagonal, simétrica y estrictamente diagonalmente dominante (diagonal $2 + h^2 > 2 =$ suma de los dos vecinos), luego SPD e invertible.

Si $u(0) = \alpha \neq 0$: solo cambia la ecuación del primer nodo interior ($i = 1$), donde $u_0 = \alpha$ se mueve al lado derecho:

$$(2+h^2)u_1 - u_2 = h^2 \sin(x_1) + \alpha.$$

La matriz A_h **no cambia**; solo se modifica la primera componente del vector del lado derecho (se le suma α).

(b) Elementos finitos (lineales a trozos). Forma débil: multiplicando por $v \in H_0^1(0,1)$ e integrando por partes (el término de frontera se anula porque $v(0) = v(1) = 0$):

$$a(u,v) := \int_0^1 u'v'\,dx + \int_0^1 uv\,dx = \int_0^1 \sin(x)v\,dx =: F(v), \qquad \forall v \in H_0^1(0,1).$$

Sea $V_h \subset H_0^1(0,1)$ el espacio de funciones lineales a trozos en la malla equidistante $x_i = ih$, $h = 1/N$, con base de funciones "sombrero" $\{\varphi_i\}_{i=1}^{N-1}$ ($\varphi_i(x_j) = \delta_{ij}$). El método de Galerkin busca $u_h = \sum_j c_j\varphi_j \in V_h$ tal que $a(u_h, \varphi_i) = F(\varphi_i)$ para $i = 1, \ldots, N-1$, es decir

$$(K+M)\mathbf{c} = \mathbf{F}, \qquad K_{ij} = \int_0^1 \varphi_i'\varphi_j' dx, \quad M_{ij} = \int_0^1 \varphi_i\varphi_j dx, \quad F_i = \int_0^1 \sin(x)\varphi_i\,dx.$$

Con las fórmulas estándar para funciones sombrero equidistantes:

$$K = \frac{1}{h}\begin{pmatrix} 2 & -1 & & \\ -1 & 2 & -1 & \\ & \ddots & \ddots & \ddots \end{pmatrix} \ (\text{matriz de rigidez}), \qquad M = \frac{h}{6}\begin{pmatrix} 4 & 1 & & \\ 1 & 4 & 1 & \\ & \ddots & \ddots & \ddots \end{pmatrix} \ (\text{matriz de masa}).$$

El sistema resultante $(K+M)\mathbf{c} = \mathbf{F}$ es también tridiagonal, SPD.

Tasa de convergencia: como $a(\cdot,\cdot)$ es coerciva y acotada en $H_0^1(0,1)$ (coerciva incluso en toda la norma H^1, gracias al término $\int uv$), el lema de Céa da

$$\|u-u_h\|_{H^1} \le C \inf_{v_h\in V_h} \|u-v_h\|_{H^1} \le C'\, h\, \|u''\|_{L^2},$$

usando la estimación estándar de interpolación lineal a trozos. Como $u\in H^2(0,1)(porque \sin(x)\in L^2$ y el problema es elíptico regular), se obtiene convergencia **de orden 1 en la norma de energía** H^1: $\|u-u_h\|_{H^1} = O(h)$. Por el argumento de dualidad de Aubin-Nitsche, la convergencia en norma L^2 es **de orden 2**: $\|u-u_h\|_{L^2} = O(h^2)$.